

**Breaking Borders: Barriers to Starting and Sustaining an International Business in
Bangladesh – A Case Study on TSMC**

Submitted to:

Limuza Fairuz

Lecturer

Army Institute of Business Administration (AIBA), Sylhet.

Course Name: International Business Environment

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Group J Members:

Md. Saiful Islam Tanmoy (101022007)

Dhrubo Barua Dip (101022027)

Sayaba Alam (101022037)

Tamima Mumi (101022070)

Tanvir Hassan Emon (101022046)

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Subject: Submission of Research Proposal on "**Breaking Borders: Barriers to Starting and Sustaining an International Business in Bangladesh – A Case Study on TSMC**"

Respected Ma'am,

We, the undersigned students of Group J, are pleased to submit our research proposal for the International Business Environment course (BBA 10), titled "**Breaking Borders: Barriers to Starting and Sustaining an International Business in Bangladesh – A Case Study on TSMC.**"

The purpose of this study is to critically examine the opportunities and challenges faced by **Taiwan Semiconductor Manufacturing Company (TSMC)**, a leading international firm, in entering the Bangladeshi market. Our objective is to analyze Bangladesh's readiness for high-tech FDI and develop actionable strategies to overcome identified market, regulatory, and infrastructural barriers. The paper incorporates comparative analysis with India and Vietnam to contextualize Bangladesh's competitiveness.

This research was conducted over three weeks using secondary data analysis from global reports, academic journals, and industry databases. We have developed a targeted investment proposal and supporting visual analyses to make the paper comprehensive and decision-oriented.

We kindly request that you accept this proposal for evaluation and look forward to your feedback on how we can further improve our understanding of real-world international business dynamics.

Thank you for your guidance throughout this assignment.

Sincerely,

On behalf of Group J.

Executive Summary

This paper examines the viability of Taiwan Semiconductor Manufacturing Company (TSMC), the world's leading semiconductor foundry, entering the Bangladeshi market. It identifies and examines the primary obstacles to Bangladeshi companies expanding internationally, with a focus on issues related to regulations, infrastructure, finances, and markets. The research emphasizes Bangladesh's distinct position as a developing yet unprepared participant in the global value chain through international comparisons, particularly with Vietnam and India. Customized suggestions are put forth to assist TSMC in overcoming these obstacles through public-private collaboration, regulatory changes, strategic alliances, and infrastructure and education investments. Last but not least, an investment plan centered on Bangladesh is offered to show how TSMC might gain an early-mover advantage while advancing the country.

Introduction

The Rise of International Business in Emerging Markets

Because it allows businesses to access new markets, lessen their reliance on a single economy, and take advantage of comparative advantages in labor, capital, and technology, international business is essential to global economic integration. Multinational corporations (MNCs) have emerged as significant actors in influencing innovation networks and trade flows in the age of globalization. In addition to increasing GDP, luring foreign companies to developing nations like Bangladesh is essential for fostering infrastructural development, knowledge sharing, industry diversification, and integration into worldwide supply chains.

Bangladesh as a Business Destination

Over the last ten years, Bangladesh has continuously maintained a GDP growth rate between 6 and 7%, making it one of the South Asian economies with the fastest rates of growth. The nation has potential for foreign investment due to its youthful population, expanding middle class, and advantageous location between South and Southeast Asia. Its global business environment is still developing, though. Its capacity to draw in high-value businesses like semiconductors has been hampered by issues like energy shortages, complicated regulatory frameworks, infrastructure constraints, and a lack of high-tech personnel. Although industries like remittances and textiles have prospered, Bangladesh has not yet completely embraced the high-tech and digital economy.

Additionally, recent government initiatives—like the creation of Special Economic Zones (SEZs), tax breaks for high-tech businesses, and the automation of administrative procedures—indicate an increasing tilt in policy toward luring knowledge-intensive foreign direct investment. However, regional neighbours like Vietnam and India continue to outperform in terms of overall ease of doing business, innovation readiness, and infrastructure capacity.

Objective and Scope

The purpose of this paper is to assess the viability of Taiwan Semiconductor Manufacturing Company's (TSMC) expansion into Bangladesh. TSMC is the world leader in contract chip manufacturing. It specifically lists the main obstacles that TSMC would have to overcome to establish and continue operations in the nation. Economic indicators, regulatory frameworks, infrastructural quality, market circumstances, talent availability, and global

competitiveness are just a few of the variables that are covered in the study. In order to make Bangladesh a more feasible and alluring location for semiconductor investment, the paper makes strategic recommendations based on comparisons with nations such as Vietnam and India. Its ultimate goal is to put Bangladesh in the global high-tech value chain and add to the policy discussion on industrial upgrading.

Organizational Overview: TSMC

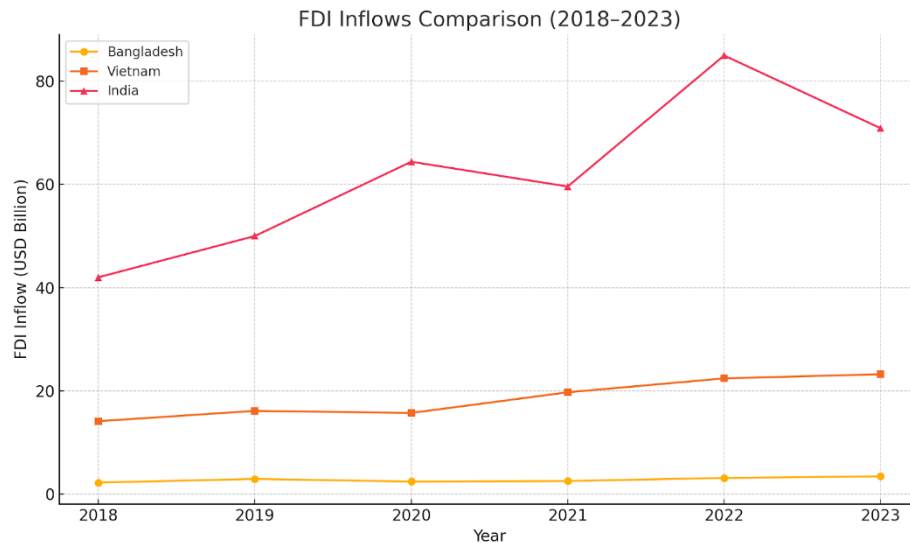
With over 50% of the global market, Taiwan Semiconductor Manufacturing Company (TSMC) is the biggest and most sophisticated specialized semiconductor foundry in the world. TSMC, which was founded in 1987, invented the foundry model, in which businesses develop chips but contract out their production. Leading tech companies like Apple, NVIDIA, Qualcomm, and AMD rely on it to power everything from high-performance computing and smartphones to AI and automotive applications. With significant fabrication facilities (fabs) in Taiwan, the United States, Japan, and Europe, TSMC is well-known throughout the world for its state-of-the-art process technologies, which include 5nm and 3nm nodes.

TSMC is expanding its manufacturing base in light of the growing geopolitical concerns in the Taiwan Strait and its strategic importance in the global supply chain. Despite being out of the ordinary, considering Bangladesh as a possible growth location may provide long-term benefits, such as cheap labor costs, a prime location, and first-mover access to South Asia's unexplored chip market.

Understanding the International Business Landscape in Bangladesh

With a population of more than 170 million and consistent economic growth, Bangladesh is still a small participant in international foreign direct investment (FDI). From 2018 to 2023, the nation received \$2 to \$3 billion in foreign direct investment (FDI) annually, which is far less than the average of \$15 billion for Vietnam and \$60 billion for India. This disparity draws attention to both systemic issues and unrealised potential, particularly in high-tech industries like semiconductors, where Bangladesh is currently underrepresented. The following comparison chart shows the inflows of foreign direct investment from 2018 to 2023.

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(Appendix B)

Economic Indicators

Manufacturing accounted for around 20% of Bangladesh's GDP in 2024, which came to about \$460 billion. Nonetheless, fewer than 1% of all trade is made up of high-tech exports, suggesting a significant reliance on low-value industries like clothing and textiles. Although the nation enjoys advantageous labour prices, its competitiveness in technology-intensive industries is limited globally due to a lack of value-added output. Though growth is still uneven, efforts are being made to diversify exports and draw investment through Hi-Tech Parks and Special Economic Zones (SEZs).

Legal and Regulatory Environment

Inconsistent policy execution, bureaucratic red tape, and slow enforcement plague Bangladesh's judicial system. Industrial licenses might take months to get. Moreover, lax enforcement of intellectual property rights (IPR) deters high-tech investors like TSMC.

Cultural, Political, and Infrastructural Context

In Transparency International's 2023 Corruption Perceptions Index, Bangladesh came in at number 149, indicating persistent problems with the government. Despite the business community's general openness to international endeavors, operational hazards include centralized decision-making and political instability. The nation's infrastructure is beset by inadequate transportation logistics, limited broadband coverage, and frequent power outages, particularly outside of major areas. According to the 2023 Global Competitiveness Report,

Bangladesh lags far behind Vietnam and India in terms of infrastructure, digital capability, and logistics.

Challenges in Market Entry and Operations

- Lack of clusters specific to a sector: no specific industrial zones for electronics or semiconductors.
- Skill shortages: There aren't enough high-tech engineers or chip designers.
- Unreliable utilities: Regular interruptions in the water and electricity supplies.
- Weak innovation ecosystem: There is little cooperation between industry, R&D facilities, and universities.

These difficulties indicate obvious areas for transformation, even while they also provide actual obstacles. Bangladesh may develop into a promising high-tech investment destination with focused regulatory support, infrastructure improvements, and strategic international alliances, particularly for businesses like TSMC looking to diversify regionally.

Key Barriers to International Business in Bangladesh

Bangladesh has a potential but undeveloped environment for doing business internationally, particularly in high-tech industries like semiconductors. Although its macroeconomic stability, cost-effective labor, and strategic location seem alluring, there are a number of operational and structural obstacles that could prevent businesses like Taiwan Semiconductor Manufacturing Company (TSMC) from entering the market and remaining there. These major issues are described in this section from the perspectives of regulations, infrastructure, culture, markets, finance, law, and industry.

Regulatory Barriers

The regulatory environment in Bangladesh is still complicated and disjointed, especially for high-tech sectors. In order to obtain land purchase, environmental clearance, tax registration, and industrial permits, TSMC would have to go through a number of lengthy bureaucratic processes that may take several months. Companies are forced to deal with several government departments in the absence of a comprehensive "one-stop service" for investors, which delays implementation and raises entry costs.

Comparison:

Through the Enterprise Law (2015) and Investment Law (2020), Vietnam updated its FDI regulations, streamlining processes, implementing digital submission platforms, and cutting licensing time frames by about 30%. Bangladesh has not yet enacted similar reforms, which have made the investment climate less flexible.

Infrastructure Limitations

- **Power Supply:** To maintain the integrity of nanometer-scale chip manufacture, TSMC's wafer fabrication plants, or "fabs," need constant, round-the-clock electricity with no voltage variation. Frequent power outages and uneven voltage are major hazards in Bangladesh, particularly outside of Dhaka.
- **Transport Infrastructure:** The nation's primary seaport, Chattogram Port, is overworked and devoid of deep-draft and container automation capabilities. Higher logistics expenses and more shipment delays result from this. The flow of raw materials and completed goods is slowed down by inland transportation's limited expressways and traffic congestion.
- **Digital Infrastructure:** In terms of digital connectivity, Bangladesh lags behind its neighbours in this area. Data-intensive businesses like semiconductors would encounter challenges in global communication, cloud services, and process automation due to the average internet speed of 34.1 Mbps (as opposed to 83.7 Mbps in India and 68.4 Mbps in Vietnam). (see **Appendix**).

Cultural and Language Barriers

Technical training resources, research articles, and employee manuals are frequently unavailable in English or current formats, despite the fact that English is widely utilized in corporate circles. Furthermore, TSMC's organized, efficiency-driven corporate approach may be in contrast to Bangladesh's company culture, which favours informal negotiation and hierarchical decision-making. Communication style misalignment may lead to conflict in local operations or joint ventures.

Market Entry Challenges

The fundamental ecosystem required for a semiconductor value chain is absent from Bangladesh. There are no well-established groups of suppliers, cleanroom makers, or chip designers. Talent availability is impacted by the lack of microelectronics, VLSI design, and

semiconductor physics courses offered by local colleges. Setting up joint ventures or technical collaborations is made more difficult by the dearth of local partners with semiconductor experience.

Comparison: With the help of international R&D facilities, startup accelerators, and engineering schools, India has established flourishing chip design hubs in places like Bangalore and Hyderabad that Bangladesh does not yet have.

Financial Barriers

For international investors, Bangladesh's financial environment presents serious risks:

- **Exchange Rate Volatility:** The Bangladeshi Taka is subject to abrupt devaluations, which may raise operating expenses and have an impact on TSMC's ability to repatriate profits.
- **High Interest Rates:** Long-term investment in capital-intensive sectors like semiconductors is restricted by high borrowing rates.
- **Limited Investment Insurance:** Bangladesh lacks comprehensive risk coverage, which makes it more difficult to protect against financial or political risks than nations that offer Export Credit Agencies (ECAs) or public-private investment guarantees.

Legal and Compliance Issues

- **IPR Protection:** Since TSMC works with sensitive proprietary technologies, lax enforcement of intellectual property rights is a major concern.
- **Dispute Resolution:** Arbitration is discouraged, and corporate operations become unclear when legal proceedings take five years or longer to conclude.
- **Rigidity of Labour Law:** Bangladesh's labour laws are more stringent than those of Vietnam with regard to overtime caps, pay scales, and hiring/firing procedures, which makes it difficult to expand operations in a flexible manner.

Barriers Specific to TSMC

The following challenges are especially pertinent to the production of semiconductors:

- **Water Supply:** Millions of litres of ultra-pure water are needed daily for the water-intensive fabrication processes. Bangladesh suffers from poor municipal infrastructure and water contamination, particularly outside of major cities.
- **Skilled Workforce:** There are no local candidates for highly specialized positions like semiconductor metrologists, wafer process analysts, and lithography engineers. Hiring such talent would necessitate either substantial local training or overseas hiring.
- **Cleanroom Standards:** Bangladesh's industrial zones do not currently support or codify the stringent environmental controls required for semiconductor manufacture (ISO 14644 Class 1–10 cleanrooms).

Strategies to Overcome Barriers

Bangladesh needs to adopt a multifaceted approach that tackles issues with infrastructure, human capital, and regulations in order to draw in a tech behemoth like TSMC. The following tactics seek to establish a favourable climate for foreign investment in high-tech:

Policy Recommendations

- **Adopt a Semiconductor Investment Act:** This specific law should ensure government-supported R&D funding, duty-free imports of capital equipment, simplified factory registration, and long-term tax breaks (10–15 years). Clear rules for IP protection and environmental compliance that are unique to semiconductor operations should also be included in this Act.
- **Create a Semiconductor Special Economic Zone (SEZ):** Bangladesh should set aside land for an SEZ specialized in the production of electronics and chips in a strategic area, ideally close to a port or international airport. High-speed internet, clean water, uninterrupted energy, and customs facilitation must all be available in this area.
- **Reforming Regulations to Make Doing Business Easier:** Establish a digital platform that serves as a one-stop shop for factory registration, FDI approvals, and compliance monitoring. Reduce processing times and boost investor trust by adopting regulatory simplification akin to Vietnam's Enterprise Law.

Public–Private Partnerships (PPPs)

- **Power and Utilities:** Construct a specialized gas or solar-powered generating unit close to the TSMC factory in partnership with regional energy providers. Reliability of power is essential for wafer manufacturing.
- **Telecommunications Infrastructure:** Install fibre-optic connections and make sure the SEZ is prepared for 5G. Meeting the data transfer requirements for chip design and logistics systems will require collaboration with telecom carriers (such as Robi and Grameenphone).
- **Water Management:** A lot of water is used in TSMC's operations. To address industrial-grade clean water needs, government and commercial partners must work together to invest in sustainable water treatment, recycling, and supply infrastructure.

Innovation and Technology Adaptation

- **Import Core Infrastructure:** Work with TSMC to import waste recycling infrastructure, sophisticated HVAC systems, and cleanroom technologies from already-existing factories in Arizona or Taiwan. This would guarantee adherence to industry standards and cut down on setup time.
- **Phased Investment through Modular Design:** Use a modular approach to factory setup rather than a full-scale foundry up front. This enables risk reduction and scalability while evaluating viability in the Bangladeshi setting.
- **Upgrading Digital Infrastructure:** To meet the digital demands of semiconductor logistics, testing, and design verification, make investments in regional data centers, cloud storage, and cybersecurity frameworks.

International Collaboration Models

- **Co-Investment in Skill Development:** Establish skill development centers in partnership with the World Bank, Asian Development Bank, and Japan International Cooperation Agency (JICA). These organizations might also offer financial support and technological know-how.

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- **Examine and Imitate International Best Practices:** Take note of semiconductor regulations from nations such as Vietnam, India, and Malaysia, paying special attention to export facilitation plans, skill certification schemes, and public-private cooperation.
- **Bilateral Agreements:** To ease TSMC's entry and guarantee security and protection for foreign investment, Bangladesh may establish bilateral trade or investment treaties with Taiwan or the United States.

Capacity Building and Skill Development

- **Semiconductor Training Centre:** established a nationwide Semiconductor Training Institute in collaboration with the National University of Singapore (NUS) and the Bangladesh University of Engineering and Technology (BUET). Courses in automated systems, fabrication, and chip design would be available at this facility.
- **Scholarship Programs:** Establish government-funded scholarships to study semiconductor physics and microelectronics in the United States, Taiwan, or South Korea for Bangladeshi students. In order to serve in the local semiconductor industry, including TSMC facilities, recipients would have to return.
- **Workforce Reskilling:** Provide technicians and mid-level employees with vocational training programs that are tailored to the particular needs of cleanroom operations and semiconductor testing settings.

Investment Proposal and Solutions Specific to TSMC

Bangladesh needs to provide a deliberate, phased investment environment that strikes a balance between opportunity and reasonable risk in order to draw in a high-value investor like TSMC. This section presents a customized plan that satisfies TSMC's operational needs and supports Bangladesh's objectives for economic growth.

Proposed Investment Model:

Phase 1: Pilot Phase (Year 1–2)

- **Investment:** \$100 million
- **Facility Type:** Advanced packaging and testing unit (back-end semiconductor operations)

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- Employment: Approximately 300 workers, focused on assembly, quality assurance, and logistics
- Objectives:
 - Test local infrastructure and regulatory processes
 - Build initial workforce capacity through on-the-job training
 - Establish local vendor and logistics partnerships

Phase 2 (Year 3–5)

- Investment: \$400 million
- Facility Type: Front-end wafer fabrication using 65nm process technology
- Employment: 1,500+ skilled engineers and technicians
- Target Market: Export-focused, supplying mid-range chips to India, the Middle East, and Southeast Asia
- Objectives:
 - Build scalable foundry operations in South Asia
 - Establish regional supply chain links for packaging, testing, and logistics
 - Serve the growing demand in developing markets for automotive, telecom, and consumer electronics

Location Recommendation:

Kaliakoir Hi-Tech Park, Gazipur

- Advantages:
 - Proximity to Dhaka (Skilled labour and administrative centre)
 - Existing infrastructure: Reliable power, road access, and partial utility readiness

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- Government incentives: Export processing zone benefits, reduced tax rates, land leasing support
- Ongoing World Bank-supported digital infrastructure projects

Comparative Market Analysis

Factor	Bangladesh	Vietnam	India
Cost of Labour	Very Low	Low	Moderate
Infra Readiness	Moderate	High	High
Regulatory Clarity	Low	Moderate	High
Talent Availability	Low	Moderate	High
FDI Incentives	Moderate	High	High

Though it lacks expertise and infrastructure, Bangladesh makes a compelling argument for first-mover investments due to its unexplored market potential and labour cost advantage. Strategic alliances and focused reforms can help alleviate the relative deficiencies.

TSMC's Rationale for Bangladesh Entry:

- Geopolitical Diversification: Bangladesh provides political neutrality and strategic diversification in Asia amid escalating tensions between China and Taiwan.
- Untapped South Asian Market: First-mover access to the Bay of Bengal Rim, which includes underserved markets like Nepal, Myanmar, and Sri Lanka, can be gained by establishing an early presence in Bangladesh.
- Export Hub Potential: Bangladesh's closeness to India and ASEAN nations provides logistical advantages for chip exports, while its LDC status (until 2026) grants special trade conditions with the EU.

- Government Support Signals: Future policy alignment is more likely given Bangladesh's policy direction under Digital Bangladesh and Smart Bangladesh, which shows a readiness to interact with high-tech industries.

Conclusion

- Bangladesh is at a turning point in its development into a force to be reckoned with in the global business arena. The nation has significant obstacles that discourage high-tech investments, including regulatory bottlenecks, infrastructural constraints, and a lack of qualified technical manpower, despite its strong economic growth and advantageous geographic location. These factors presently outweigh the advantages of reduced operating costs and market potential for multinational leaders like TSMC.
- These difficulties are not insurmountable, though. Bangladesh may implement specific legislative changes, expedite investment processes, and make significant investments in STEM education and workforce development by taking inspiration from regional success stories like Vietnam and India. The core capabilities required to sustain a semiconductor ecosystem can be developed through strategic public-private partnerships and international cooperation, especially in cleanroom technology, research and development, and technical training.
- The possible arrival of TSMC might revolutionize the market. It would signal a calculated move away from reliance on remittances and textiles and toward participation in the global knowledge economy. In addition to creating jobs, this kind of investment would stimulate long-term industrial diversification, infrastructure growth, and innovation.
- Bangladesh must move quickly and cooperatively to achieve this goal, coordinating international collaborations, private sector preparedness, and government incentives. The country can genuinely transcend boundaries and become a vibrant center for the global high-tech industry if it adopts a proactive, forward-thinking strategy.

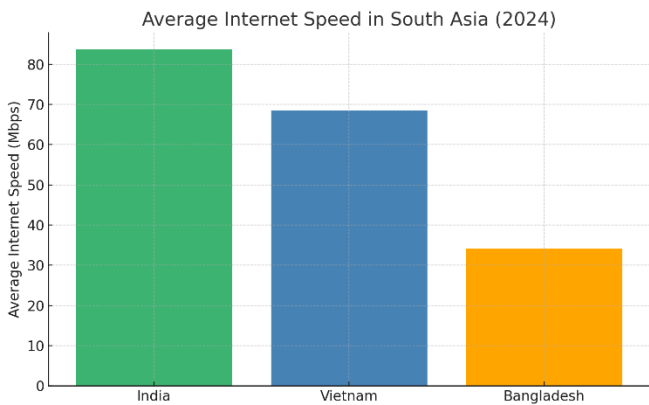
References

1. World Bank. (2024). *World Development Indicators*.
<https://databank.worldbank.org/source/world-development-indicators>
2. UNCTAD. (2024). *World Investment Report 2024*.
<https://unctad.org/publication/world-investment-report-2024>
3. Taiwan Semiconductor Manufacturing Company (TSMC). (2023). *Annual Report 2023*.
<https://investor.tsmc.com/english/annual-reports>
4. Transparency International. (2023). *Corruption Perceptions Index 2023*.
<https://www.transparency.org/en/cpi/2023>
5. World Economic Forum. (2023). *The Global Competitiveness Report 2023*.
<https://www.weforum.org/reports/global-competitiveness-report-2023>
6. Bangladesh Investment Development Authority (BIDA). (2024). *Investment Guide*.
<https://bida.gov.bd/>
7. Government of Vietnam. (2015). *Vietnam Enterprise Law 2015*.
<https://vbpl.vn/botuphap/Pages/vbpqen-toanvan.aspx?ItemID=11063>
8. Ookla(2024). *Speedtest Global Index*.
<https://www.speedtest.net/global-index>

Appendices

Appendix A: Internet Speed Comparison Chart

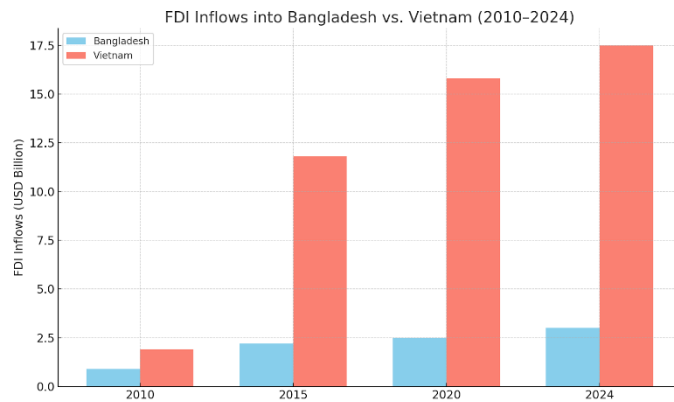
Chart: Internet Speed in South Asia (2024)



Country	Avg Speed (Mbps)
India	83.7
Vietnam	68.4
Bangladesh	34.1

Source: Ookla Speedtest Index

Appendix B: FDI Inflows (Bangladesh vs Vietnam)



Source: UNCTAD, World Bank

The following graph contrasts FDI inflows between Bangladesh, Vietnam, and India during the previous six years (2018–2023) to give a better idea of how well Bangladesh has performed in luring FDI.

